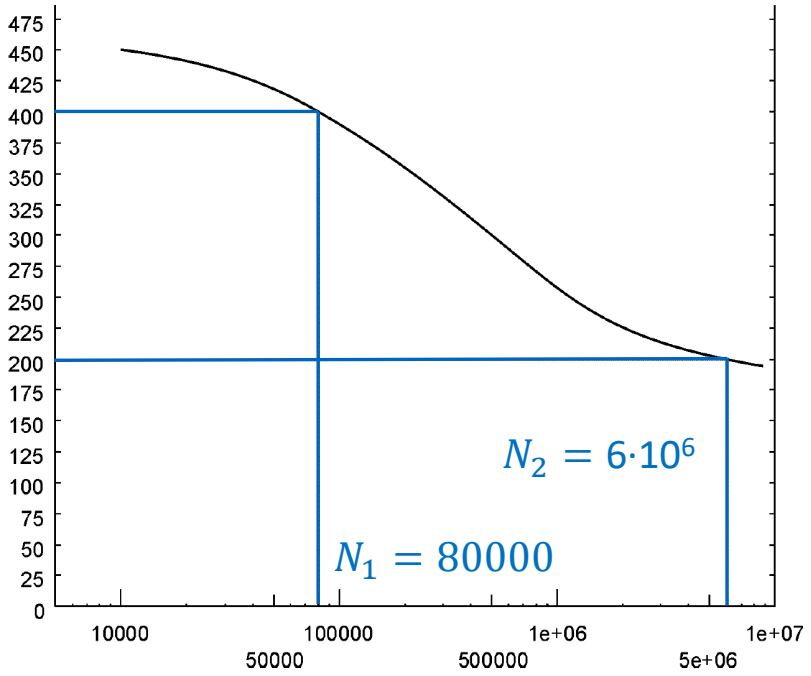
A part made of a material with the characteristic S-N curve shown in the figure below is subject to $n_1 = 80000$ load cycles at a stress amplitude of $\sigma_1 = 400MPa$ and $n_2 = 6{\cdot}10^6$ cycles at an amplitude of $\sigma_2 = 200MPa$.

Task 1: According to the Miner rule, by how much must the number of cycles be reduced to avoid failure?

$$\sum_i \frac{n_i}{N_i} = \frac{n_1}{N_1} + \frac{n_2}{N_2} = 2.0$$

$$N_z \underbrace{\sum_i \frac{n_i}{N_i}}_{2.0} = 1 \quad \Longrightarrow \quad \boldsymbol{N_z = \frac{1}{2}}$$

The damage according to the miner Miner rule is 200% per cycle.
Therefore, the cycle number has to be reduced by 0.5



Task 2: What is the maximum stress amplitude, so that another part, made of the same material, can withstand n = 100 000 cycles using a factor of safety of 5?

$$\frac{n}{N} = 0.2$$

$$N = \frac{n}{0.2} = 500000$$

$$\sigma_a = 300 \text{ MPa}$$

